Educational 3-Phase PCB BLDC Motor Control System

System Architecture, Electromagnetic Modeling, Sensing & Driver Implementation Specification

1. Executive Summary & System Overview

This document details the engineering specifications for an educational planar PCB brushless DC (BLDC) motor platform designed for laboratory instruction in embedded motor control, electromagnetic commutation, and power electronics. The platform interfaces a custom stator PCB with an Arduino UNO R4 Minima microcontroller, three dedicated MP6550 H-bridge motor drivers with integrated current sensing, an array of ratiometric analog Hall effect sensors, and an onboard synchronous DC-DC step-down converter.

System Parameter	Specification Value	Engineering Rationale & Notes
Stator Architecture	6 Planar PCB Coils (3-Phase, Concentric/Spiral)	2 coils per phase connected in series/parallel pairs (60° mechanical spacing).
Rotor Architecture	8 Permanent Magnets (Alternating N-S)	4 pole pairs ($p = 4$), resulting in 90° mechanical per electrical period.
Pitch Circle Diameter (D_{pitch})	$\varnothing 35.0\text{ mm}$ ($R_{pitch} = 17.5\text{ mm}$)	Shared centerline diameter for magnets, coil active tracks, and sensors.
Magnet Dimensions & Air Gap	$\varnothing 10.0\text{ mm} \times 1.0\text{ mm}$ (NdFeB N35/N42), $4.0 - 5.0\text{ mm}$ Air Gap	Yields $\approx \pm 35\text{ mT}$ to $\pm 50\text{ mT}$ flux density at sensor active layer.
Phase Coil Resistance	$17.84\,\Omega$ per driven phase pair	Limits coil current to $\approx 140.1\text{ mA}$ at $2.5\text{

System Parameter	Specification Value	Engineering Rationale & Notes
		V}$ bus supply.
Main Driver Supply Voltage (V_{MOT})	2.5 V DC (via Synchronous Buck)	Derived from primary 5.0 V rail; guarantees safe thermal operation on PCB traces.
Controller & Drivers	Arduino UNO R4 Minima + 3x MP6550 ICs	Renesas RA4M1 48 MHz ARM Cortex-M4 with 14-bit ADC and high-side current sense.

2. Magnetic Geometry & Sensor Field Calculations

The permanent magnet rotor utilizes eight axially magnetized neodymium disc magnets ($\varnothing 10\text{ mm} \times 1\text{ mm}$, $B_r \approx 1.2\text{ T}$). The magnetic flux density $B_z(z)$ along the central axis of a cylindrical magnet at distance z is governed by:

```
B_z(z) = (B_r / 2) * [ (z + t) / sqrt((z + t)^2 + R^2) - z / sqrt(z^2 + R^2) ]
```

Where $B_r = 1.2\text{ T}$, magnet thickness $t = 1.0\text{ mm}$, and magnet radius $R = 5.0\text{ mm}$. At an operating distance $z = 4.5\text{ mm}$ (accounting for rotor clearance and PCB thickness to bottom layer), the calculated on-axis field is approximately 42.5 mT (425 Gauss). As the rotor rotates, the tangential flux modulation experienced by the underlying sensors approximates a sinusoid with peak amplitude $\hat{B} \approx \pm 40\text{ mT}$.

3. Analog Hall Sensor Selection

Analog (linear) Hall effect sensors output a voltage proportional to the applied perpendicular magnetic field, allowing continuous rotor angle estimation rather than the coarse 60° resolution provided by digital latching sensors.

Sensor Part Number	Package Type	Sensitivity ($V_{CC} = 5\text{V}$)	Linear Range	Evaluation for this Setup
SS49E / AH49E	TO-92S / SOT-23	$1.40 - 1.80\text{ mV/Gauss}$ ($14 - 18\text{ mV/mT}$)	$\pm 65\text{ mT}$ ($\pm 650\text{ G}$)	**Optimal for Labs:** Readily available, low cost, full dynamic range utilized without saturation ($V_{OUT} \approx 1.8\text{V} - 3.2\text{V}$).
DRV5055A4 (TI)	SOT-23 / TO-92	12.5 mV/mT	$\pm 85\text{ mT}$	**High Precision:** Superior thermal stability and low noise; ratiometric output perfectly tracks 5V rail.
A1308KUA-2-T (Allegro)	SIP-3 / SOT-23W	2.50 mV/Gauss (25 mV/mT)	$\pm 80\text{ mT}$	**Alternative:** Excellent linearity; factory-trimmed quiescent voltage offset.

4. Spatial Placement & Mathematical Angle Estimation

The motor features $N_s = 6$ stator coils and $2p = 8$ rotor poles (pole pairs $p = 4$). The fundamental relationship between electrical angle θ_e and mechanical angle θ_m is:

```
θ_e = 4 * θ_m   ==>   1 Electrical Period (360°e) = 90° Mechanical
```

Recommended 3-Phase Symmetric Sensor Placement (Layout B)

To maximize instructional symmetry and maintain direct compatibility with both sinusoidal field orientation (FOC) and 6-step trapezoidal commutation, three linear sensors (H_U, H_V, H_W) are placed along the $\varnothing 35.0\text{ mm}$ pitch circle at 30° mechanical intervals (120° electrical

shift):

- **Sensor H_U (0° Mechanical):** Aligned directly beneath the center radial axis of Stator Coil 1 (Phase U).
- **Sensor H_V (30° Mechanical):** Positioned symmetrically in the inter-coil gap between Coil 1 and Coil 2 (120° electrical offset).
- **Sensor H_W (60° Mechanical):** Aligned directly beneath the center radial axis of Stator Coil 2 (Phase V) (240° electrical offset).

Clarke Transform & Angle Calculation

Let V_U, V_V, V_W be the zero-centered, normalized analog voltages from the three sensors:

```
α = V_U
β = (1 / √3) * (V_U + 2 * V_V)   [or β = (V_V - V_W) / √3]

θ_e = atan2(β, α)
θ_m = θ_e / 4  (+ sector_offset * 90°)
```

5. DC-DC Converter & Power Distribution

Each phase pair exhibits a DC resistance of $R_{coil} = 17.84\,\Omega$. Supplying the motor bridge with $V_{MOT} = 2.5\text{ V DC}$ ensures controlled current draw and prevents thermal over-stressing of the PCB copper traces.

- **Single Phase Peak Current:** $I_{phase} = 2.5\text{ V} / 17.84\,\Omega = 140.13\text{ mA}$
- **Dual Active Phase Maximum Current:** $I_{total, peak} = 2 \times 140.13\text{ mA} = 280.26\text{ mA}$
- **Output Power at 2.5V:** $P_{out} = 2.5\text{ V} \times 0.280\text{ A} = 0.70\text{ W}$
- **Input Current from 5V Supply ($\eta \approx 90\%$):** $I_{in, 5V} = 0.70\text{ W} / (5.0\text{ V} \times 0.90) = 155.6\text{ mA}$ (Well within the 0.75 A limit)

Converter Solution	Topology / Part Number	Key Electrical Parameters	Form Factor & Usage
Integrated IC	Texas Instruments **TLV62569DBVR**	$2.5\text{V} - 5.5\text{V}$ Input, 1.5A Output, 1.5 MHz Sync Buck	SOT-23-5; high efficiency ($>92\%$), low component count (1 inductor, 2 ceramic caps).

Converter Solution	Topology / Part Number	Key Electrical Parameters	Form Factor & Usage
Adjustable Module	**MP1584EN Mini Buck Board**	$4.5\text{V} - 28\text{V}$ Input, Up to 3A Output, Adjustable Pot	$17\text{ mm} \times 22\text{ mm}$ board; trim onboard pot to exactly 2.50 V.
Fixed Output Module	Pololu **D24V5F2 / D24V10F2**	$3.0\text{V} - 36\text{V}$ Input, 2.5V Fixed, $500\text{mA} - 1\text{A}$	Breadboard-compatible 3-pin SIP header; eliminates calibration errors in student kits.

6. Microcontroller & Driver Interfacing

The system uses an **Arduino UNO R4 Minima** (Renesas RA4M1 32-bit ARM Cortex-M4) paired with three monolithic **MP6550** H-bridge motor drivers.

ADC Configuration & Reference Voltage

- **Analog Reference:** Configure the Arduino to AR_DEFAULT (5.0 V). Because ratiometric Hall sensors scale output linearly with V_{CC}, powering both sensor V_{CC} and the ADC reference from the same 5.0 V rail provides automatic common-mode noise and power supply rejection.
- **ADC Resolution:** Enable 14-bit sampling mode (analogReadResolution(14)) to achieve 0.305 mV per LSB resolution.

MP6550 Driver & Current Sense Configuration

- **Control Lines:** Each MP6550 driver controls one motor phase using two PWM pins (IN1, IN2) for complementary switching or sign-magnitude modulation.
- **Current Limit ($ISET$):** Set the MP6550 current limit resistor R_{ISET} to allow up to 300 mA:
  `R_ISET = 200kΩ * (100mA / I_LIMIT) ==> R_ISET ≈ 66.5kΩ`
- **Current Feedback:** Connect the analog current monitor outputs (V_{ISET}) of the three drivers to Arduino analog pins A3, A4, and A5 to monitor per-phase current dynamics.

7. Complete Arduino UNO R4 Implementation Code

```
/*
  Educational 3-Phase PCB BLDC Motor Controller
  Target: Arduino UNO R4 Minima (Renesas RA4M1)
  Sensing: 3x Linear Hall Sensors (HU=A0, HV=A1, HW=A2)
  Drivers: 3x MP6550 H-Bridges (Phase U: D3/D5, Phase V: D6/D9, Phase W: D10/D11)
*/

#include <Arduino.h>

const int PIN_HALL_U = A0;
const int PIN_HALL_V = A1;
const int PIN_HALL_W = A2;

const int PIN_U_IN1 = 3;
const int PIN_U_IN2 = 5;
const int PIN_V_IN1 = 6;
const int PIN_V_IN2 = 9;
const int PIN_W_IN1 = 10;
const int PIN_W_IN2 = 11;

// Calibration Offsets (Quiescent 2.5V center at 14-bit ADC = 8192)
int offset_U = 8192;
int offset_V = 8192;
int offset_W = 8192;

const float SQRT3 = 1.7320508f;
const int POLE_PAIRS = 4;

void setup() {
  Serial.begin(115200);

  // Configure ADC Reference and 14-bit Resolution
  analogReference(AR_DEFAULT);      // 5.0V ratiometric reference
  analogReadResolution(14);         // 0 to 16383 count range

  // Driver PWM Pin Configuration
  pinMode(PIN_U_IN1, OUTPUT); pinMode(PIN_U_IN2, OUTPUT);
  pinMode(PIN_V_IN1, OUTPUT); pinMode(PIN_V_IN2, OUTPUT);
  pinMode(PIN_W_IN1, OUTPUT); pinMode(PIN_W_IN2, OUTPUT);

  // Set initial driver state to high-impedance
  setPwmDrive(0, 0, 0);

  // Optional Auto-zero calibration routine
```

```
  calibrateHallSensors();
}

void calibrateHallSensors() {
  long sumU = 0, sumV = 0, sumW = 0;
  for (int i = 0; i < 256; i++) {
    sumU += analogRead(PIN_HALL_U);
    sumV += analogRead(PIN_HALL_V);
    sumW += analogRead(PIN_HALL_W);
    delayMicroseconds(500);
  }
  offset_U = sumU / 256;
  offset_V = sumV / 256;
  offset_W = sumW / 256;
}

void loop() {
  // 1. Sample Analog Hall Sensors
  float u = (float)(analogRead(PIN_HALL_U) - offset_U);
  float v = (float)(analogRead(PIN_HALL_V) - offset_V);
  float w = (float)(analogRead(PIN_HALL_W) - offset_W);

  // 2. Forward Clarke Transform (αβ)
  float alpha = u;
  float beta = (u + 2.0f * v) / SQRT3;

  // 3. Electrical Angle Estimation
  float theta_e = atan2(beta, alpha); // Returns angle in radians [-PI, PI]
  if (theta_e < 0) theta_e += 2.0f * PI;

  // 4. Mechanical Angle Calculation
  float theta_m = (theta_e / (float)POLE_PAIRS) * (180.0f / PI);

  // 5. Example Sinusoidal Commutation Output (PWM duty generation)
  float advance_angle = theta_e + (PI / 2.0f); // 90 deg lead for torque
  int pwm_u = (int)(127.0f + 100.0f * sin(advance_angle));
  int pwm_v = (int)(127.0f + 100.0f * sin(advance_angle - 2.0f * PI / 3.0f));
  int pwm_w = (int)(127.0f + 100.0f * sin(advance_angle + 2.0f * PI / 3.0f));

  setPwmDrive(pwm_u, pwm_v, pwm_w);

  // Telemetry Output
  static unsigned long lastPrint = 0;
  if (millis() - lastPrint > 50) {
    lastPrint = millis();
```

```
    Serial.print("Theta_e_deg:"); Serial.print(theta_e * 180.0f / PI);
    Serial.print("\tTheta_m_deg:"); Serial.println(theta_m);
  }
}

void setPwmDrive(int u, int v, int w) {
  analogWrite(PIN_U_IN1, u); analogWrite(PIN_U_IN2, 0);
  analogWrite(PIN_V_IN1, v); analogWrite(PIN_V_IN2, 0);
  analogWrite(PIN_W_IN1, w); analogWrite(PIN_W_IN2, 0);
}
```

8. Pedagogical Laboratory Curriculum Outline

1. ---

   Lab 1: Static Magnetic Field Characterization & ADC Calibration
   Students manually rotate the rotor, record raw ADC values from H_U, H_V, H_W, calculate offset/gain mismatches, and verify sinusoidal flux distribution.
2. **Lab 2: Vector Transformations & Clarke Transformation**
   Implement real-time Clarke transformation on the Arduino UNO R4 Minima. Plot the resulting trajectory on the α-β plane using the Arduino IDE Serial Plotter to evaluate circle circularity and eccentricity.
3. **Lab 3: 6-Step Trapezoidal Commutation with Dynamic State Estimation**
   Construct a state-machine comparator based on continuous Hall signals to generate digital Hall equivalents, evaluating torque ripple and back-EMF spikes.
4. **Lab 4: Closed-Loop Field-Oriented Control (FOC) & Current Feedback**
   Utilize the MP6550 current feedback pins with PI current regulation loops to implement smooth continuous torque and velocity control.